Aditya Tripathy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.69
Delhi Public School, Vasant Kunj	Class XII (CBSE)	2024	96.00%
Delhi Public School, Vasant Kunj	Class X (CBSE)	2022	96.40%

SCHOLASTIC ACHIEVEMENTS

Insitute Academic Excellence Award

- Awarded by the **Director of IIT Hyderabad** and the **Principal Adviser to the Prime Minister** for **Highest CGPA (10.0)** in the incoming Batch of 2028.

POSITIONS OF RESPONSIBILITY

Overall Head, Kludge - *Cybersecurity and Networking Club*

- Designing introductory learning material and CTF problems based on Cryptography and Binary Exploitation for the Batch of 2029.
- Managing a team of 13 cores and mentoring in the topics of Binary Exploitation and Reverse Engineering.

Core, Kludge - *Cybersecurity and Networking Club*

- Designed CTF challenges based on **Cryptography**, **Web Security**, **Binary Exploitation**, **Reverse Engineering** and **Forensics** for **MilanCTF** and **EnigmaCTF** and helped manage the smooth conduction of the event with over 250+ participants.
- Conducted sessions on Cybersecurity topics for the IITH community.

Teaching Assistant for **EE1010 (Mathematics for Electrical Engineers)**

- Corrected answer scripts for tests and helped in exam invigilation.

PROJECTS

Dynamically Morphable RISC-V CPU | (Github)

- **Architected** a 5-stage pipelined RISC-V CPU with full support for RV32I and RV64I instruction sets, integrating **hazard detection**, **data forwarding**, and 1-bit **branch prediction**.
- **Engineered** a runtime-configurable architecture toggled via a dedicated mode bit, enabling the CPU to **dynamically morph** between a **single 64-bit core** (for **data intensive** tasks like matrix multiplication) and **dual parallel 32-bit cores** (for **control heavy** tasks like graph traversal).
- **Achieved** an **8.36% improvement** in **performance per watt** compared to a static 64-bit baseline by optimizing execution time and dynamic power consumption across unified and split modes.
- **Deployed** and validated the custom processor architecture using a block design on an **AMD PYNQ-Z2 FPGA.**

Asynchronous Kernel I/O Framework in xv6 | (Github)

- **Architected** an ***io_uring style*** asynchronous I/O subsystem within the RISC-V **xv6 operating system kernel**, integrating thread safe, **ring-buffered submission and completion queues** to offload blocking tasks to a background worker pool.
- **Engineered** a **context-bypass mechanism** enabling **global kernel threads** to map process-local **file descriptors**, manage **concurrent inode locks**, and execute safe **cross address space memory** copies via copyout to stream disk data directly into user buffers.
- **Validated** system resilience and absolute memory isolation under **multi-process concurrency** using a custom sliding window stress testing suite**,** achieving **100% data integrity** across **800 high frequency reads** while maintaining **zero vulnerability** to malicious pointer injections.

Flipper Zero Compatible Multi-Tool PCB | (Github)

- **Engineered** a custom **mixed-signal PCB** architecture inspired by the Flipper Zero, built around the **STM32WB55RG wireless microcontroller** to drive Bluetooth (BLE), an LCD interface, and tactile inputs.
- **Designed** schematics and high-density PCB layouts in **Altium Designer**, integrating a **CC1101 Sub-GHz module** with custom **impedance-matched RF traces** tightly optimized for **433MHz operation**
- **Simulated** and validated high-speed digital and **RF signal integrity** (SI) using **Ansys SIWave**, successfully restricting return loss (S11 parameters) to strictly under the -20dB constraint for both the 433MHz and Bluetooth antennas.

SKILLS

- **Programming Languages**: C/C++, Python, Go, Bash
- **Tools:** Make, CMake, gdb, Git/Github, LaTeX, Docker, AMD Vivado, Intel Quartus, Ansys SIWave
- **Libraries**: NumPy, Matplotlib, SciPy, seaborn

RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Data Structures & Applications, Computer Architecture, Machine Learning
- **Mathematics**: Probability, Random Processes, Introduction to Statistics, Matrix Theory, Differential Equations & Transform Techniques, Vector Calculus
- **Electrical Engineering:** Signal Processing, Control Systems, PCB Design, Digital Systems, Analog Electronics, Semiconductor Device Fundamentals